

EXPERIMENT 25:

Demonstrate Infrastructure as a Service (IaaS) by creating a resources group using a Public Cloud Service Provider (Azure), configure with minimum CPU, RAM, and Storage.

Aim

To demonstrate Infrastructure as a Service (IaaS) using Microsoft Azure by creating a Resource Group and deploying a lightweight virtual machine with minimum available CPU, RAM, storage, and networking resources.

Procedure

1. Open the Microsoft Azure Portal and sign in using the Azure account.
2. Search for Resource Groups, select Create, choose the required subscription and region, and create a Resource Group named IaaS-Practical-RG.
3. Search for Virtual Machines, select Create → Azure Virtual Machine, and select the previously created Resource Group.
4. Enter the virtual machine name as IaaS-Test-VM and select Ubuntu Linux as the operating system image.
5. In the Size section, select the smallest suitable VM available. From the available options, select B2ats_v2, which provides 2 vCPUs and 1 GiB RAM.
6. In the Disks section, select the minimum suitable OS disk and do not add additional data disks. Configure the required virtual network, subnet, public IP, and network security group under Networking.
7. Configure an administrator username and SSH authentication, review all settings, and click Create to deploy the virtual machine.
8. After deployment, open the VM's Overview page and verify the Resource Group, VM size, operating system, CPU, RAM, storage, and running status.
9. Connect to the Ubuntu VM using SSH and verify the resources using commands such as `lscpu`, `free -h`, and `df -h`.
10. Take screenshots of the Resource Group, VM configuration, selected B2ats_v2 size, and resource verification commands as evidence of the implementation.

IMPLEMENTATION:

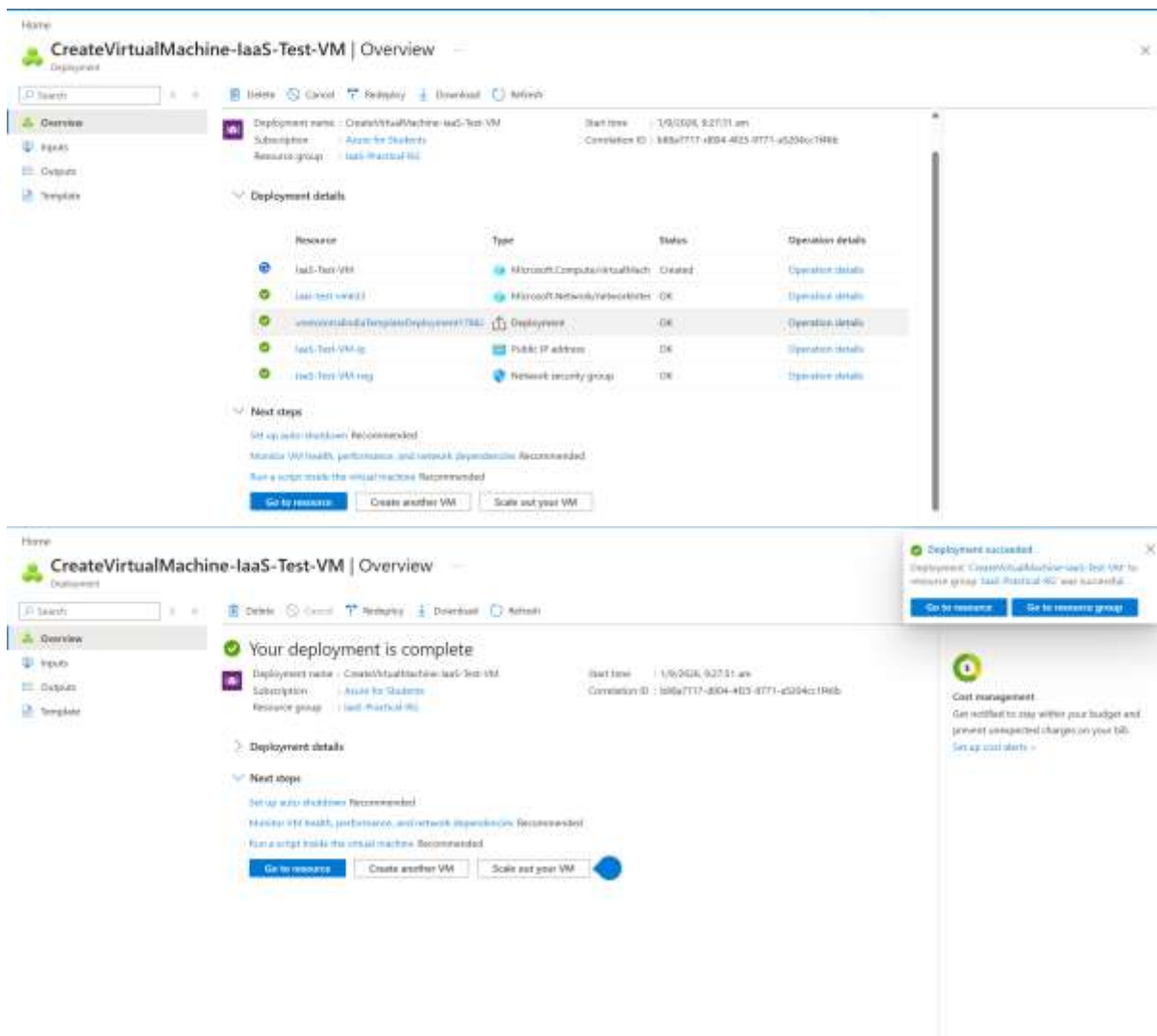
Home > Compute infrastructure | Virtual machines

Create a virtual machine

*** Initializing deployment...
Initializing template deployment to resource group 'IaaS-Practical-RG'.

BasicsDisksNetworkingManagementMonitoringAdvancedReview + create

Login with Microsoft Entra ID	Off
Backup	Off
Site Recovery	Off
Enable periodic assessment	Off
Enable hotpatch	Off
Patch orchestration options	Image default
Auto-shutdown	Off
Enable hibernation	Off
Monitoring	
Alerts	Off
Boot diagnostics	On
Enable OS guest diagnostics	Off
Enable application health monitoring	Off
Advanced	
Extensions	None
VM applications	None



Result

The IaaS environment was successfully created using Microsoft Azure. A Resource Group and Ubuntu Virtual Machine were deployed with the minimum available configuration of 2 vCPUs, 1 GiB RAM, and minimum suitable storage. The virtual machine and its allocated resources were successfully verified through the Azure Portal and Linux system commands.